

Pre-Lab 11: Bitwise Basics (Preparation for Labwork 11)

Focus: Bit positions, masks, and simple bitwise operations on 32-bit unsigned integers.

General Rules

- Use 32-bit unsigned integers.
 - Bits are numbered from the right starting at 0 (LSB).
 - When counting from the left (MSB side), bit positions are 31, 30, 29, ... down to 0.
 - Unless the question says "modify in-place", return the new value.
 - Do not use loops or arrays unless the question explicitly requires them.
 - All return values for "get" questions must be exactly 0 or 1.
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Questions

P1. Bit Index Check (Right Side)

Write a function that returns the value of the 0th bit (LSB) of an unsigned integer.

- Input: an unsigned integer x
- Output: 0 or 1

P2. Single-Bit Mask

Write a function that returns a number where only the 12th bit from the right is 1 and all other bits are 0.

- Input: none
- Output: an unsigned integer with exactly one bit set
- Clarify: 12th bit from the right means bit position 11

P3. Test a Bit (Right Side)

Write a function that returns 1 if the 10th bit from the right is set, otherwise 0.

- Input: an unsigned integer x
- Output: 1 if bit 9 is 1, otherwise 0

P4. Clear a Bit (Right Side)

Write a function that clears the 3rd bit from the right to 0 and returns the new value.

- Input: an unsigned integer x
- Output: a new unsigned integer with bit 2 forced to 0

P5. Toggle Two Bits

Write a function that toggles both the 1st and 4th bits from the right and returns the new value.

- Input: an unsigned integer x
- Output: a new unsigned integer with bits 0 and 3 flipped

P6. Extract a 4-Bit Field

Write a function that returns the 4-bit value stored in bits 4..7 (inclusive) of an unsigned integer.

- Input: an unsigned integer x
- Output: an unsigned integer in the range 0..15
- Clarify: bits 4..7 means positions 4, 5, 6, 7

P7. Set a 4-Bit Field

Write a function that replaces the 4-bit field in bits 8..11 with a value v (0..15).

- Input: an unsigned integer x and an unsigned integer v (0..15)
- Output: a new unsigned integer with bits 8..11 set to v

P8. MSB Side Check

Write a function that returns the value of the 2nd bit from the left.

- Input: an unsigned integer x
- Output: 0 or 1
- Clarify: 2nd bit from the left means bit position 30

P9. MSB Side Clear

Write a function that clears the 1st bit from the left to 0 and returns the new value.

- Input: an unsigned integer x
- Output: a new unsigned integer with bit 31 forced to 0

P10. Even/Odd Without Modulus

Write a function that returns 1 if the number is odd and 0 if it is even, using only bitwise operations.

- Input: an unsigned integer x
- Output: 1 for odd, 0 for even